Chemistry Topic 1 — Atomic Structure and Periodic Classification

Current Affairs (2024–2026 Focus)

- **IUPAC Superheavy Elements & Island of Stability:** Synthesizing superheavy transactinide elements in the 7th period: **Nihonium (Z=113, Nh)**, **Moscovium (Z=115, Mc)**, **Tennessine (Z=117, Ts)**, and **Oganesson (Z=118, Og)**. Research targeting Z=119 (Ununennium) and Z=120 in the hypothetical 8th period to confirm Glenn Seaborg's predicted 'Island of Stability' (Z=114, N=184).
- **Rare Earth Elements (REEs) & Critical Minerals Mission:** India's Ministry of Mines identified **30 critical minerals** (including Lithium, Cobalt, Nickel, and the 17 Rare Earth Elements: 15 Lanthanides + Scandium + Yttrium). IREL operating monazite processing facilities in Odisha and Kerala; geological discoveries of Lithium reserves in Reasi (Jammu & Kashmir) and Degana (Nagaur, Rajasthan).
- **Gallium and Germanium Global Export Controls:** Strategic semiconductor metalloids. Gallium (Z=31) and Germanium (Z=32) are critical for military radar electronics (Gallium Nitride, GaN), high-efficiency satellite photovoltaics, and fiber-optic infrared optics.
- **National Quantum Mission & Trapped-Ion Atoms:** Development of neutral atom quantum processors utilizing Rydberg states of Group 1 alkali metals (**Rubidium-87, Cesium-133**) and alkaline earth ions (**Strontium-88, Ytterbium-171**).
- **International Year of the Periodic Table (IYPT):** UNESCO commemorated the 150th anniversary of Dmitri Mendeleev's periodic system (1869).

1.1 Structure of the Atom: Classical Models to Quantum Wave Mechanics

Evolution of Atomic Models

- **Dalton's Atomic Theory (John Dalton, 1808):** Matter consists of indivisible, indestructible atoms. Atoms of the same element are identical in mass and properties. (Overturned by the discovery of subatomic particles and isotopes).
- **Thomson's Plum Pudding Model (1898):** Uniform sphere of positive charge with electrons embedded like raisins in a pudding.
- **Rutherford's Planetary Nuclear Model (1911):** Discovery of the nucleus. Failed to explain why accelerating orbiting electrons do not continuously radiate electromagnetic energy and spiral into the nucleus within 10^{-8} s (Maxwell's electrodynamics).
- **Bohr's Quantum Postulates (1913):** Introduced non-radiating stationary orbits where angular momentum is quantized: $mvr = nh/2\pi$. Successfully explained the hydrogen spectral emission lines (Lyman, Balmer, Paschen, Brackett, Pfund series).
- **De Broglie's Matter Waves (1924):** $\lambda = h/p = h/mv$. Dual particle-wave nature of electrons.

- **Heisenberg's Uncertainty Principle (1927):** \Delta x \cdot \Delta p \ge h / 4\pi. Impossible to determine simultaneous trajectory; replaced deterministic orbits with probabilistic 3D **Orbitals**.

1.2 Quantum Numbers, Shells and Electronic Configuration

Orbital Quantum Numbers & Shapes

- **Principal (n = 1, 2, 3, 4):** Shell designation (K, L, M, N). Total orbitals in shell = n^2; Max electrons = 2n^2.
- **Azimuthal (l = 0\text{ to } n-1):** Subshell and shape.
- l = 0 (s orbital): Spherical, non-directional; can hold 2 electrons.
- l = 1 (p orbital): Dumbbell-shaped (p_x, p_y, p_z); can hold 6 electrons.
- l = 2 (d orbital): Double-dumbbell (d_{xy}, d_{yz}, d_{zx}, d_{x^2-y^2}, d_{z^2}); can hold 10 electrons.
- l = 3 (f orbital): Complex multi-lobed; 7 orbitals holding 14 electrons.
- **Rules Governing Electron Filling:**
- **Aufbau Principle:** Orbitals fill in order of increasing (n + l) energy: 1s < 2s < 2p < 3s < 3p < 4s < 3d < 4p < 5s < 4d < 5p < 6s < 4f < 5d < 6p < 7s.
- **Pauli Exclusion:** Max 2 electrons per orbital with opposing spins.
- **Hund's Rule:** Degenerate orbitals fill singly with parallel spins before pairing.

1.3 Atomic Number, Mass Number, Isotopes and Isobars

Nuclear Classification Summary

- **Atomic Number (Z):** Number of nuclear protons.
- **Mass Number (A):** Sum of protons and neutrons (Z + N).
- **Isotopes:** Same Z, different A (^1_1H, ^2_1H, ^3_1H).
- **Isobars:** Same A, different Z (^{40}*{18}Ar and ^{40}*Ca).
- **Isotones:** Same neutron count (N = A - Z).
- **Isoelectronic:** Same total electron count (O^{2-}, F^-, Na^+, Mg^{2+}).

1.4 Evolution of the Periodic Table: Mendeleev to Moseley

Historic Milestones

- **Mendeleev's Table (1869):** Arranged 63 elements by **atomic weight**. Predicted Gallium (Ga) as *Eka-Aluminum* and Germanium (Ge) as *Eka-Silicon*.
- **Moseley's Modern Law (1913):** Shifted basis to **Atomic Number (Z)**, eliminating anomalies. Long form table has **7 Periods and 18 Groups**.

1.5 Master Analysis of Periodic Trends and Anomalies

Core Trends Across Period vs Down Group

- **Atomic Radius:** Decreases across period \to; increases down group \downarrow.
- **Ionization Energy (IE):** Increases across period \to; decreases down group \downarrow.
- *Anomalies:* Be > B (full 2s^2) and N > O (half-filled 2p^3).
- **Electron Affinity (EA):** Increases across period \to; decreases down group \downarrow.
- *Anomaly:* **Chlorine has higher affinity than Fluorine** (Cl > F > Br > I) due to inter-electronic repulsions in tiny fluorine.
- **Electronegativity (EN):** Increases across period \to; decreases down group \downarrow. Peak is **Fluorine (4.0)**.

1.6 Metals, Non-Metals, Metalloids and Body Composition

Chemical Demarcation

- **Metals:** Electropositive, form basic oxides, malleable, ductile, high electrical conductivity.
- **Non-Metals:** Electronegative, form acidic oxides, brittle insulators (except graphite).
- **Metalloids:** Borderline semiconductors: B, Si, Ge, As, Sb, Te, Po.
- **Elemental Abundance:**
- Earth's Crust: Oxygen (46.6\%) > Silicon (27.7\%) > Aluminum (8.1\%, most abundant metal) > Iron (5.0\%) > Calcium (3.6\%).
- Human Body: Oxygen (65\%) > Carbon (18.5\%) > Hydrogen (9.5\%) > Nitrogen (3.2\%) > Calcium (1.5\%, most abundant metal).

Bilingual Terminology -- Atomic Structure and Periodic Table

English Term	Hindi Term	Key Anchor
Atom (परमाणु)	परमाणु	Smallest particle of element retaining chemical properties
Atomic Number (परमाणु क्रमांक)	परमाणु संख्या	Number of protons; unique identifier of element
Atomic Mass (परमाणु द्रव्यमान)	परमाणु द्रव्यमान	Protons + neutrons (roughly); measured in amu

English Term	Hindi Term	Key Anchor
Valence Electrons (संयोजकता इलेक्ट्रॉन)	संयोजी इलेक्ट्रॉन	Electrons in outermost shell; determine chemical behaviour
Periodic Table (आवर्त सारणी)	आवर्त सारणी	Arrangement of elements by atomic number; Mendeleev (1869)
Period (आवर्त)	आवर्त	Horizontal row of periodic table; 7 periods
Group (समूह)	समूह / वर्ग	Vertical column; 18 groups; same valence electrons = similar properties
Metals (धातु)	धातु	Left and centre of table; conductors, lustrous, malleable
Non-metals (अधातु)	अधातु	Right side; insulators; brittle; e.g., O, N, S, Cl, P
Metalloids (उपधातु)	उपधातु	Properties between metals and non-metals; Si, Ge, As
Noble Gases (उत्कृष्ट गैस)	अक्रिय गैस / उत्कृष्ट गैस	Group 18; completely filled shells; extremely unreactive
Alkali Metals (क्षार धातु)	क्षार धातु	Group 1 (Li, Na, K, Rb, Cs, Fr); 1 valence e-; very reactive
Alkaline Earth Metals (क्षारीय मृदा धातु)	क्षारीय मृदा धातु	Group 2 (Be, Mg, Ca, Sr, Ba, Ra); 2 valence e-
Halogens (हैलोजन)	हैलोजन	Group 17 (F, Cl, Br, I, At); 7 valence e-; very reactive
Transition Metals (संक्रमण धातु)	संक्रमण तत्त्व	Groups 3-12; d-block; variable valency; coloured compounds
Electronegativity (विद्युत-ऋणात्मकता)	विद्युत-ऋणात्मकता	Ability to attract electrons in bond; increases across period (right); decreases down group

One-Liner Revision -- Atomic Structure and Periodic Table

#	Fast Fact
1	Mendeleev's Periodic Table (1869) = arranged by **atomic mass**; predicted missing elements
2	Modern Periodic Table = arranged by **atomic number** (Moseley, 1913)
3	Periodic table: **7 periods** (horizontal rows) + **18 groups** (vertical columns)
4	Group 1 = Alkali metals (H, Li, Na, K, Rb, Cs, Fr); highly reactive
5	Group 17 = Halogens (F, Cl, Br, I, At); 7 valence electrons; most electronegative
6	Group 18 = Noble gases (He, Ne, Ar, Kr, Xe, Rn); inert; completely filled shells
7	Most electronegative element = **Fluorine (F)**
8	Largest atom = Cesium (Cs) / Francium (Fr) [by atomic radius]
9	Smallest atom = Helium (He) / Hydrogen (H)

#	Fast Fact
10	Hardest natural substance = **Diamond** (form of Carbon)
11	Lightest element = **Hydrogen (H)**; atomic number 1
12	Lightest metal = **Lithium (Li)**; also most reactive metal
13	Heaviest naturally occurring element = **Uranium (U)**; atomic number 92
14	Most abundant element in universe = **Hydrogen**
15	Most abundant element in Earth's crust = **Oxygen** (46%)
16	Most abundant metal in Earth's crust = **Aluminium** (Al; 8%)
17	Most abundant gas in atmosphere = **Nitrogen** (78%)
18	Electronic configuration rule: 2n^2 (max electrons per shell: 2, 8, 18, 32...)
19	Octet rule: atoms tend to gain/lose electrons to have 8 valence electrons
20	Isotopes = same Z, different A; e.g., H-1 (protium), H-2 (deuterium), H-3 (tritium)
21	Isobars = same A, different Z; e.g., Ar-40 and Ca-40
22	Isotones = same number of neutrons, different Z; e.g., C-13 and N-14
23	Radioactive elements: all elements with Z > 83 (Bismuth) are naturally radioactive
24	Lanthanides = f-block elements 57-71; Actinides = f-block elements 89-103
25	Transition metals = d-block (Groups 3-12); variable oxidation states; coloured ions
26	Metalloids (semiconductors): B, Si, Ge, As, Sb, Te, At
27	Allotropy = element existing in multiple forms; Carbon (diamond, graphite, fullerene)
28	Diamond = hardest; Graphite = soft, good conductor; both are pure carbon
29	Polonium = only element with all naturally occurring isotopes being radioactive
30	Element 118 (Oganesson) = heaviest known element; all artificial/synthetic